

Task 1:-

Dominant Kernel :- Binary 1D Convolution
(second convolution layer)

Tool :- torch, Conv 1D

C-IN = 64 , C-OUT = 64 , k = 3 , L = 500

Weight precision : 1-bit packed Binary

Activation precision : INT8 (1 byte per value)

Operation : XOR + pop count (replaces multiply-accumulate)

Task 2:-

$$\text{FLOPs} = 2 \times \text{C-IN} \times \text{C-OUT} \times k \times L$$

$$= 2 \times 64 \times 64 \times 3 \times 500$$

$$= 12,288,000 \text{ FLOPs.}$$

Task 3:-

Reverse pattern :- Weight - stationary

(Weights loaded once, activations stream thru for all positions)

LOWER BOUND: - No reuse:

$$\text{Weights} = 64 \times 64 \times 3/8 = 1,536 \text{ bytes}$$

$$\text{Inputs} = 64 \times 500 \times 1 \text{ byte} = 32,000 \text{ bytes}$$

$$\text{Outputs} = 64 \times 500 \times 1 \text{ byte} = 32,000 \text{ bytes}$$

$$\text{Total} = 65,536 \text{ bytes}$$

UPPER BOUND - full weight reuse:

$$\text{Weights on-chip (not loaded)} = 0 \text{ bytes}$$

$$\text{Inputs} = 64 \times 500 \times 1 \text{ byte} = 32,000 \text{ bytes}$$

$$\text{Outputs} = 64 \times 500 \times 1 \text{ byte} = 32,000 \text{ bytes}$$

$$\text{Total} = 64,000 \text{ bytes}$$

Task 4:-

$$\text{AI - Lower} = 12,288,000 / 65,536 = 187.5 \text{ flops/byte}$$

$$\text{AI - upper} = 12,288,000 / 64,000 = 192.0 \text{ flops/byte}$$

Task 5.1

Q 1.1

Bottle neck = Compute units.

Both AI Bounds (187.5 and 192 flops/byte) are far from to the right of the sky130 ridge pt (16 flops/byte) and the max ridge pt (33.3 flops/byte). The Kernel is Compute - Bound on Both platforms.

Currently the Compute units are limited by the 43.14 ns critical path thru the xor + pop count adder tree. (from out 1.696 on inv-2 gate of from timing - report) which restricts the clock to 19.7 MHz instead of 100 MHz Target.

Q 2.1 pipeline the xor + pop count adder tree in

Compute - Core. SV across 4 stages with registered intermediate sums btw each stage, breaking the 43.14 ns critical path into four - long stages and closing timing at 100 MHz.